

Learning Matchmaking Patterns from Wedding Announcements Using TabTransformerPredicting Marriage Compatibility Using TabTransformer

Project Check-in #2

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1. Title

- Title: Learning Matchmaking Patterns from Wedding Announcements Using TabTransformer
- Devpost: <https://devpost.com/software/matchmaking-patterns-from-wedding-announcements/joins/VaqV8T-SHFGSdIbmXmaJAw>
- Github: <https://github.com/YanmiYu/DL-Marriage-Prediction>
- The main idea of this project is to learn matchmaking patterns from existing couple data to predict a likely profile of an ideal partner. This is formulated as a multi-label, multi-class classification problem.
 - Input: Structured profile of a person (e.g., gender, age, occupation, education, city)
 - Output: A set of ideal partner characteristics (e.g., preferred age range, occupation type, education level, region)

2. Group Members

Name	Login
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Yicen Ye	yyc45

3. Introduction

a. How do we arrive the marriage pattern problem and our motivation:

- Marriage patterns often reflect broader social, cultural, and economic structures. This project explores how structured personal attributes influence matchmaking patterns, using wedding announcement data from The New York Times.
- Initially the idea was motivated by the study of another research of Brown, which targets this problem more from a quantitative sociological perspective.
- Inspired by this idea, combined with the deep learning framework we learned so far, we want to build a model that, given one individual's profile, can predict the most likely profile of their ideal match. We hope this could support research into relationship preferences and social stratification, or serve as the backend for a matchmaking recommender system.
- Technically more specifically, we were inspired by recent work on deep learning for tabular data and saw an opportunity to explore a social application of TabTransformer, a Transformer-based architecture for structured data.

b. Problem Type

- This is a multi-output classification problem: each output field (e.g., partner's age range, job category, etc.) is treated as a separate classification task.

4. Related Work

In exploring modeling approaches for structured data beyond our primary use of TabTransformer (Huang et al., 2020), we also reviewed *TabNet: Attentive Interpretable Tabular Learning* by Arik and Pfister (2021). TabNet introduces a sequential attention mechanism that selects meaningful feature subsets at each decision step, allowing for both high performance and interpretability in tabular data modeling. Compared with TabNet, the architecture of TabTransformer enhances the modeling of categorical tabular features by combining trainable embeddings and self-attention mechanisms, allowing the model to learn dependencies between feature columns—such as how education level may relate to occupation or geography. Although our project focuses on TabTransformer, understanding TabNet helped us contextualize our work within the broader landscape of neural models for structured data and appreciate different design approaches to handling feature interactions in tabular inputs.

URLs :

- TabTransformer paper: <https://arxiv.org/abs/2012.06678>
- TabNet: <https://arxiv.org/abs/1908.07442>
- PyTorch implementation by lucidrains:
<https://github.com/lucidrains/tab-transformer-pytorch>
- Alternative: [RTDL \(Yandex Research\)](#)

5. Data

a. Dataset Description

We are using a dataset of 9,160 wedding announcements published in *The New York Times* between 2012 and 2023. The data was collected using the NYT Article Search API, and each article was processed via GPT-4 and rule-based methods to extract structured demographic features for both individuals in a couple.

b. Data Source

The data is annotated and cleaned by Brown sociologist Dr. Zhenchao Qian and senior research scientist Guixing Wei at the Data Science Initiative (DSI).

Dataset link: [x data_2013_to_2023.xlsx](#)

c. Features/Records

Over 60 structured variables were extracted, including:

- Gender
- Age (binned into ranges such as 25–30, 30–35)
- Occupation (mapped to standard categories)
- Education level
- Race
- Region
- And many other social and biographical attributes
- Size & Preprocessing

Total records: **9,160 couples** (18,320 individuals)

d. Preprocessing Steps

- Handle missing values and noisy text caused by OCR errors
- Standardize categorical fields (e.g., occupation, education, and location)
- Bin numerical fields like age into discrete ranges
- Filter to include only **heterosexual couples** (for 1:1 matching simplicity)

e. Announcement Identification Criteria

The following heuristic rules were used to classify an article as a wedding announcement:

- **2012–2017:**
 - Section = "Fashion & Style"
 - Subsection = "Weddings"
- **2017–2020:**
 - Subsection = "Weddings", or
 - Headline kicker includes "*vows*" or "*mini vows*"
- **2020–2023:**
 - Headline kicker includes "*vows*" or "*mini vows*"

f. Note on Bias

This dataset is not fully representative of the general population. It reflects both self-selection bias (only couples who submit announcements) and editorial bias (NYT tends to feature highly educated, affluent, often East Coast individuals). These biases must be acknowledged when analyzing model outcomes and generalizing findings.

6. Methodology

a. Model Architecture

- **Input Encoding:**
 - Categorical variables (e.g., gender, job category, education level, region):
 - Each category is mapped to a unique trainable embedding vector. This allows the model to learn a dense representation of each category that captures its semantic similarity to others.
 - Numerical variables (e.g., age):
 - We discretize age into bins (e.g., 18–25, 26–30, etc.) and treat them as categorical variables. Each bin is also embedded, preserving ordinal information while aligning with the model's embedding-based structure.
- **Transformer Encoder:**
 - Each feature is treated as a token. These tokens are passed into a multi-layer Transformer Encoder, which applies multi-head self-attention to capture interactions between all features (e.g., how gender and occupation jointly influence preferred partner attributes).
 - We do not use positional encodings, since feature order is fixed and not meaningful.
 - Attention layers learn interactions between features
- **Output Heads:**
 - For each target partner attribute (e.g., B_age_range, B_edu_level, etc.), we attach a separate classification head:
 - A fully connected (Dense) layer
 - One softmax layer per target (multi-output classification)
 - E.g., 5-class softmax for B_age_range

b. Model Training

- **Loss Function:** compute the cross-entropy loss for each individual output head. The total loss is the sum of all individual losses, optionally weighted to account for class imbalance or differing field importance.
- **Optimizer:** Adam (Temporary)

- **Learning Rate Scheduling:** We may apply a cosine decay schedule or a warm-up + decay strategy to stabilize early training and enhance convergence speed.
- **Regularization & Batching:**
 - Apply dropout within the Transformer encoder layers to prevent overfitting. (and maybe compare results w/o dropout)
 - Train with standard mini-batch gradient descent.
- **Training Duration (Temporary):** Training is expected to run for 10–30 epochs, depending on performance trends observed on a held-out validation set.

c. Backups:

If TabTransformer proves ineffective or computationally inefficient, we could try:

- TabNet: as mentioned in the related work review, it's a sequential attention-based model for tabular data that offers interpretability through feature masks.
- GNN: GNNs could learn embedding representations of each node based on the structure of the graph and their features, which may be useful for the link prediction (like marriage linking), and classification

7. Metrics

a. Success Measure

Success in our project means the model can accurately and meaningfully predict a set of ideal partner attributes given a person's demographic profile. Since this is a multi-output, multi-class classification problem (for the target goal), we evaluate both overall and per-output performance.

b. Planned Experiments

- Main experiment: Train a TabTransformer with multiple classification heads, each predicting one attribute of the ideal partner (e.g., age range, education level).
- Ablation / Sensitivity test: Evaluate how model performance changes when different combinations of input fields are included (e.g., with/without occupation).
- Ranking experiment (for stretch goal): Rank 10 candidates by predicted match score and evaluate whether true partners rank near the top (using ranking metrics such as NDCG or MAP if appropriate).

c. Metrics to Be Used

- Accuracy: For each classification head (e.g., how often the predicted education level matches the ground truth)
- Macro F1 / Weighted F1 Score: To ensure fair evaluation across imbalanced categories
- Confusion Matrix: To identify which categories are commonly confused
- (Optional for Stretch Goal): Ranking-based metrics (e.g., MRR, NDCG) if we implement pairwise scoring

d. Paper Related

We are adapting TabTransformer (Huang et al., 2020) for a novel application: predicting multiple labels simultaneously in a social context. The original paper evaluated classification tasks (e.g., income prediction) using accuracy and AUC, but they did not

explore multi-task or recommender-style evaluation. Therefore, we design our own metric strategy tailored to our unique setup.

e. Goal Settings

Goal	Description
Base Goal	• Train a TabTransformer that, given a person’s demographic profile (e.g., gender, race, education, occupation), predicts a single most likely bundle of ideal partner attributes (e.g., partner's age range, education level, region, etc.). • Should be able to train on a smaller data set (i.e, recent 3~5 years) and run successfully.
Target Goal	• Extend the model to output the top 3 or top 10 most likely bundles of ideal partner characteristics. • This allows us to evaluate prediction diversity and simulate more realistic matchmaking scenarios based on ranked outputs.
Stretch Goal	• Build an alternative prediction head or scoring logic that, given one person’s profile, assigns a compatibility score to 10 (or k) candidate profiles. • This enables pairwise matching without retraining the model—only modifying the inference layer and evaluation strategy. • Or, alternatively, collect more data (like before 2013) and train on a wider scope of the data.

8. Ethics

a. What broader societal issues are relevant to your chosen problem space?

Our project lies at the intersection of sociology, relationships, and data modeling. Matchmaking and partner preferences are deeply tied to societal structures—including class, race, education, and geography. Using machine learning to model romantic compatibility risks reinforcing existing social norms or inequalities. For example, if historically observed pairings reflect patterns of homogamy (people marrying within their race, class, or education level), our model may reproduce and normalize those preferences rather than challenge them. Additionally, applying such models to real-world matchmaking or dating tools could lead to algorithmic steering—where recommendations reinforce status-based sorting—without critical reflection.

b. What is your dataset? Are there any concerns about how it was collected, or labeled? Is it representative? What kind of underlying historical or societal biases might it contain?

Our dataset consists of approximately 9,000 wedding announcement articles published by The New York Times between 2012–2023. The data includes structured attributes about each spouse, such as occupation, education, and geographic location. However, this dataset is highly selective and not representative of the broader population.

There are two levels of bias embedded in the data:

- Self-selection bias: Only couples who choose to submit to the NYT (and are aware of the opportunity) are considered.
- Editorial bias: The NYT editorial team curates which announcements are published, often favoring individuals with elite education, high-status professions, and East Coast connections.

As a result, the dataset over-represents affluent, highly educated, and often white individuals. These structural biases need to be acknowledged when interpreting the model's results—especially if it is extended beyond this specific sociocultural context.

9. Division of labor: Briefly outline who will be responsible for which part(s) of the project.

Our team consists of four members. We have divided the project responsibilities into key roles to ensure clear ownership and efficient collaboration:

Yanmi Yu (yyu111) – Data Lead

- Responsible for understanding the dataset and executing preprocessing tasks
- Standardizes categorical fields (e.g., occupation, education, city)
- Handles feature engineering, binning, and missing value imputation
- Documents the full data pipeline for reproducibility

Mingrui Chen (mchen288) – Model Lead

- Designs and implements the TabTransformer architecture
- Builds multi-output classification heads for partner attributes
- Manages training procedures, loss function design, and optimization strategies
- Coordinates with the evaluation lead for model tuning and diagnostics

Yixiao Zhang (yzha1016) – Evaluation & Analysis Lead

- Conducts model evaluation using metrics such as accuracy, F1 score, and confusion matrices
- Performs ablation and sensitivity studies (e.g., input feature importance)
- Visualizes model outputs and interprets prediction behaviors
- Supports hyperparameter tuning and model robustness checks

Yicen Ye (yye45) – Product & Documentation Lead

- Defines base, target, and stretch goals of the project
- Leads the writing of project deliverables including outline, final report, and Devpost submission
- Prepares slides and visual materials for the project presentation
- Communicates the motivation, design, and outcomes in a coherent narrative

Team Collaboration

While we have defined individual roles for clarity and efficiency, our team worked collaboratively throughout the project.

We frequently met to brainstorm ideas (in-person and via Zoom), review progress, and make key decisions collectively. This ensured that all team members had a strong understanding of each component of the workflow and contributed meaningfully to the project as a whole.